

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record how its auditory cortex responds. We'll use SONIC as a starting point and work out what suits our mouse setup.

1. Choose the sounds

We'll make a program that lets us choose each tone's pitch, how long it plays, how loud it is, and the silence between tones. Each tone will fade in and out briefly to avoid clicks. We'll save the sequence so we know which sound was played each time.

SONIC used 10 ms tones with 2 ms fades at each end in awake sheep. Those are useful reference values. We still need to choose our mouse settings and decide whether the mouse will be awake or anesthetized.

2. Set up the speaker

We'll first check what audio equipment the lab already has. The TDT MF1 speaker and SA1 amplifier used in SONIC are one option to consider.

We'll keep the speaker in a fixed position and use a suitable calibrated microphone where the mouse's ear will be. This lets us check how loud each tone actually is.

3. Put the sound and brain recordings on one timeline

The playback device will schedule each tone together with a small electrical timing pulse. The brain recorder will save that pulse alongside the neural signals.

We'll also align the microphone recording with the neural recording to check when the sound actually reaches the ear. If we add behavior video later, we'll align its frame times with the same timeline.

Our first step is to identify the brain recorder and available audio equipment. Then we'll test the sound and timing on the bench, including unwanted electrical pickup. Once those checks work, the team can use its mouse protocol to start with tones separated by silence and see how the brain responds.

References: SONIC, bioRxiv v1, Methods 6.1–6.2 (p.11) and 6.6 (p.13); TDT speaker information; TDT timing information. This is our proposed mouse setup.